PUBLIC TRAFFIC FINES MANAGEMENT SYSTEM

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Problem Statement

Traffic enforcement generates large volumes of violation records. When violation information, fine assessments and payment status are difficult to review together, administrators may struggle to summarize violation patterns and monitor outstanding balances. This project proposes a structured Oracle database with PL/SQL routines for managing traffic violations, fine assessments and payment records, supported by an interactive dashboard.

Proposed Solution

Build an academic prototype in Oracle Database to store and manage traffic-violation records, fine assessments and payment transactions. PL/SQL will enforce validation and business rules, calculate outstanding balances, and produce reports. Microsoft Power BI will connect to the Oracle database and display interactive indicators and trends. The prototype is not an official government service and will not process real payments.

Objectives

Main: Design and implement an Oracle database application for recording and analyzing traffic violations, fine assessments and payment status.
Specific: (1) Import and clean a real public traffic-violations dataset. (2) Normalize the data to Third Normal Form (3NF). (3) Develop PL/SQL procedures, functions, cursors, packages, triggers and exception handling. (4) Demonstrate fine and payment tracking using clearly labelled simulated records where source data is unavailable. (5) Connect Power BI to Oracle for interactive reporting.

Data Source

Montgomery County, Maryland – Traffic Violations, published through Data.gov and the county open-data portal: https://catalog.data.gov/dataset/traffic-violations. The dataset contains electronic traffic-violation information, is updated daily, and excludes information that could uniquely identify the vehicle, vehicle owner or issuing officer. The county FY2026 Open Data Operations Manual identifies it as one of the county's largest datasets, with over two million rows. A documented subset will be used for the project. The public dataset does not supply this prototype's fine-payment lifecycle; any such records added for demonstration will be marked as simulated.

Planned Tables

VIOLATION_TYPE (violation categories); **LOCATION** (violation areas); **TRAFFIC_VIOLATION** (source violation event and available attributes); **FINE_ASSESSMENT** (assessment amount, date, due date and status); **PAYMENT** (payment amount, date, method and reference). Final columns and relationships will be confirmed after inspecting the source fields and completing Phase II normalization.

Planned PL/SQL Features

Procedures to insert/update violation and fine records and record payments; functions to calculate outstanding balances; cursors for violation and payment reports; packages to group fine-management operations; triggers to validate transactions and maintain audit information; and exception handling for invalid identifiers, invalid amounts, duplicate records and overpayments. Business rules will remain in PL/SQL.

Innovation

Option: Analytics dashboard. **Tool:** Microsoft Power BI connected to the Oracle database. The dashboard will present KPI cards, filters and charts for violation counts, violation categories, time/location patterns, assessed fines, payments and outstanding balances. It must use data from the working Oracle database, not static screenshots or mock-ups.

Expected Results and Limitations

Expected results are a normalized Oracle database, validated data, reusable PL/SQL program units, audit/validation rules and a working Power BI dashboard. The selected dataset represents Montgomery County, Maryland, not Rwanda. The prototype will not claim to represent Rwandan traffic laws or official fine amounts. Fine assessments and payment transactions created where needed will be simulated and labelled as such.

References (APA)

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